$x = 0$	0	0	0
$x = 1$	1	1	1
$x = 2$	10/3	10/3	10/3

$\mathbb{E}[XY \mid X]$

(a)

0	1/3	2/3
0	1	2
0	5/3	10/3

$Y\,\mathbb{E}[Y \mid X]$

(b)

1/3	1/3	1/3
3/2	3/2	3/2
3	3	3

$\mathbb{E}[Y^2 \mid X]$

(c)